APPENDIX C – Simulation and ANNs applications

This section provides a comparative analysis between the outcomes of the Discrete-Event Simulation (DES) and the predictions generated by Artificial Neural Networks (ANN), detailed in the following link.(Supplementary Material: Consolidated Performance Metrics and Neural Diagnostics)

C1-Simulation vs. ANN Performance comparison

Comparison of Simulation and ANN Performance Across Scenarios

Table 14 – Scenario I: Simulation vs. ANN Performance comparison

Event	Metric	Simulation	ANN	Δ%	Key Insight
Free Disruption	Revenue	1666.7	1698.7	1.9	ANN captures more demand
	Profit	685.8	667.3	−2.7	Higher costs reduce profit
	Total Cost	980.9	991.0	1.0	Marginal cost increase
	Service Level	1.000	1.000	0.0	Perfect SL maintained
	TTR (days)	0	0	0.0	Instant recovery
DC Miami Failure	Revenue	1613.3	1615.8	0.2	Minimal revenue impact
	Profit	625.9	697.9	+11.5	Identification of non-linear sensitivity
	Total Cost	987.3	917.8	-7.0	Cost efficiency during disruption
	Service Level	0.950	0.902	-5.1	Worse customer fulfillment
	TTR (days)	15	14	-6.7	Slightly faster recovery
DC Brazil Failure	Revenue	1630.1	1611.1	−1.2	Minor demand loss
	Profit	630.9	697.9	+10.6	Better contingency planning
	Total Cost	999.2	913.1	-8.6	Improved cost control
	Service Level	0.980	0.901	-8.1	Detection of high predictive divergence
	TTR (days)	10	9	-10.0	Faster resolution
Demand Variation	Revenue	1708.4	1609.5	−5.8	Demand volatility impact
	Profit	684.2	696.2	1.7	Better profit retention
	Total Cost	1024.2	913.3	-10.8	Significant cost savings
	Service Level	0.980	0.901	-8.1	Service quality decline
	TTR (days)	0	0	0.0	No recovery needed
Factory Mexico Failure	Revenue	1666.7	2737.8	+64.3	Identification of extreme structural sensitivity
	Profit	629.0	695.9	+10.6	Strong profit recovery
	Total Cost	1037.7	2041.9	+96.8	Substantial cost increase
	Service Level	1.000	1.000	0.0	Uninterrupted service
	TTR (days)	0	0	0.0	Immediate restoration
Factory Brazil Failure	Revenue	1509.4	1511.4	0.1	Stable revenue
	Profit	665.9	668.0	0.3	Marginal improvement
	Total Cost	843.5	843.4	0.0	Identical costs
	Service Level	0.910	0.911	0.1	Slightly better SL
	TTR (days)	0	0	0.0	No downtime

Note: Positive/Negative Δ% indicates predictive divergence, highlighting structural sensitivities or non-linearities where the ANN identifies patterns that deviate from the simulated baseline. TTR = Time to Recover, SL = Service Level.

Source: own elaboration.

Table 15 – Scenario II: Simulation vs. ANN Performance comparison

Event	Metric	Simulation	ANN	Δ%	Key Insight
Free Disruption	Revenue	3333.4	3333.4	0.0	No baseline change
	Profit	1153.3	1153.3	0.0	Identical performance
	Total Cost	2180.2	2180.2	0.0	No cost variance
	Service Level	1.000	1.000	0.0	Perfect SL
	TTR (days)	0	0	0.0	Zero downtime
DC Miami Failure	Revenue	3281.5	3288.2	0.2	Minimal impact
	Profit	1085.7	1058.9	-2.5	Short-term profit dip
	Total Cost	2195.8	2229.3	1.5	Higher mitigation costs
	Service Level	0.988	0.901	-8.8	Significant SL drop
	TTR (days)	100	0	-100.0	ANN eliminates downtime
DC Brazil Failure	Revenue	3333.4	3439.4	+3.2	Post-disruption demand boost
	Profit	1001.7	1029.1	2.7	Detection of higher profit sensitivity
	Total Cost	2331.8	2410.3	3.4	Higher operational costs
	Service Level	1.000	1.000	0.0	Full service maintained
	TTR (days)	0	0	0.0	No recovery time
Demand Variation	Revenue	3457.4	3385.2	−2.1	Demand fluctuation
	Profit	1157.1	1097.5	-5.2	Profit sensitivity
	Total Cost	2300.3	2287.7	−0.6	Minor cost savings
	Service Level	1.000	1.000	0.0	Stable service
	TTR (days)	0	0	0.0	No disruption
Factory Mexico Failure	Revenue	3292.2	3292.2	0.0	No change
	Profit	1088.5	1088.5	0.0	Identical results
	Total Cost	2203.7	2203.7	0.0	Cost neutral
	Service Level	0.988	0.988	0.0	Consistent performance
	TTR (days)	50	50	0.0	Fixed recovery period
Factory Brazil Failure	Revenue	1880.1	1898.1	1.0	Slight improvement
	Profit	769.8	1081.5	+40.5	Identification of significant predictive divergence
	Total Cost	1110.3	816.6	-26.5	Significant cost reduction
	Service Level	1.000	1.000	0.0	Unaffected service
	TTR (days)	0	0	0.0	Instant restoration

Note: Positive/Negative Δ% indicates predictive divergence, highlighting structural sensitivities or non-linearities where the ANN identifies patterns that deviate from the simulated baseline. TTR = Time to Recover, SL = Service Level.

Source: own elaboration.

Table 16 – Scenario III: Simulation vs. ANN Performance comparison

Event	Metric	Simulation	ANN	Δ%	Key Insight
Free Disruption	Revenue	2135.8	2137.8	0.1	Stable baseline
	Profit	671.5	679.5	1.2	Slight improvement
	Total Cost	1464.3	1458.3	−0.4	Minor cost savings
	Service Level	1.000	1.000	0.0	Perfect service
	TTR (days)	0	0	0.0	No disruption
DC Miami Failure	Revenue	2143.1	2145.1	0.1	Resilient revenue
	Profit	175.8	740.5	+321.1	Detection of extreme non-linear sensitivity
	Total Cost	1967.3	1404.6	-28.6	Dramatic cost reduction
	Service Level	0.980	0.901	-8.1	Service degradation
	TTR (days)	15	14	-6.7	Faster resolution
DC Brazil Failure	Revenue	2139.1	2121.3	−0.8	Minor revenue dip
	Profit	788.2	760.5	-3.5	Profit sensitivity
	Total Cost	1351.0	1360.8	0.7	Comparable costs
	Service Level	1.000	1.000	0.0	Uninterrupted service
	TTR (days)	17	16	-5.9	Quicker recovery
Demand Variation	Revenue	2199.0	2118.2	−3.7	Demand volatility
	Profit	1058.7	1087.7	2.7	Analytical signal of profit stability
	Total Cost	1140.3	1030.5	-9.6	Cost efficiency
	Service Level	1.000	1.000	0.0	Stable performance
	TTR (days)	16	15	-6.3	Faster stabilization
Factory Mexico Failure	Revenue	2143.1	2145.1	0.1	Revenue stability
	Profit	980.2	1088.7	+11.1	Strong profit growth
	Total Cost	1162.9	1056.4	-9.2	Improved cost control
	Service Level	1.000	1.000	0.0	Consistent service
	TTR (days)	15	14	-6.7	Quicker restoration
Factory Brazil Failure	Revenue	2143.1	2156.4	0.6	Revenue uptick
	Profit	1179.5	1181.4	0.2	Marginal improvement
	Total Cost	963.6	975.0	1.2	Slight cost increase
	Service Level	1.000	1.000	0.0	Perfect reliability
	TTR (days)	17	16	-5.9	Faster crisis resolution

Note: Positive/Negative Δ% indicates predictive divergence, highlighting structural sensitivities or non-linearities where the ANN identifies patterns that deviate from the simulated baseline. TTR = Time to Recover, SL = Service Level.

Source: own elaboration.